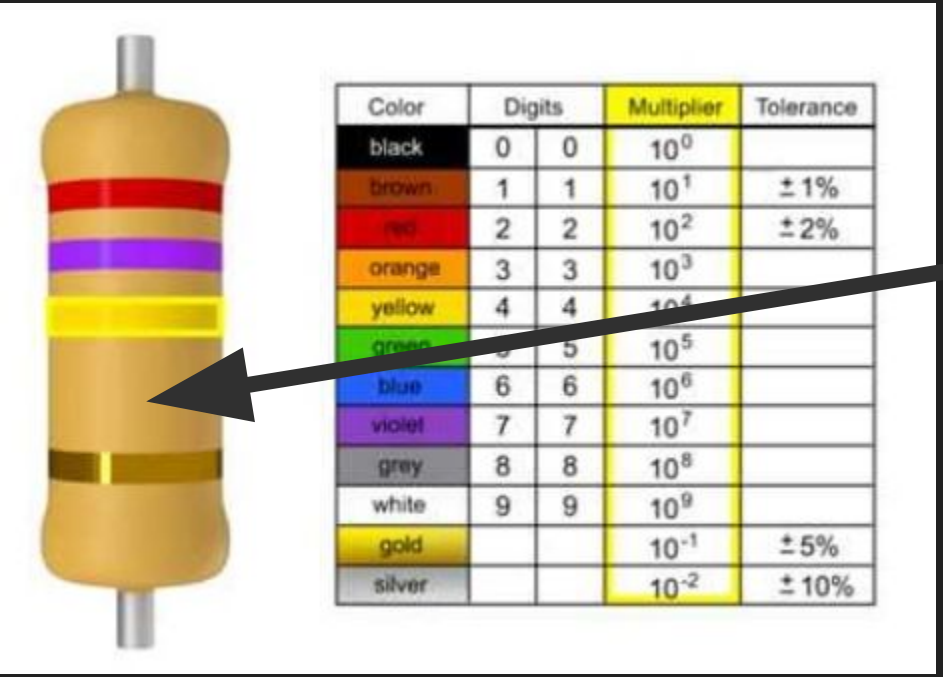
# Digital Electronics Interfacing

Lesson 3

# Resistors



| Color | Digits | | Multiplier | Tolerance |
|---|---|---|---|---|
| black | 0 | 0 | $10^0$ | |
| brown | 1 | 1 | $10^1$ | ±1% |
| red | 2 | 2 | $10^2$ | ±2% |
| orange | 3 | 3 | $10^3$ | |
| yellow | 4 | 4 | $10^4$ | |
| green | 5 | 5 | $10^5$ | |
| blue | 6 | 6 | $10^6$ | |
| violet | 7 | 7 | $10^7$ | |
| grey | 8 | 8 | $10^8$ | |
| white | 9 | 9 | $10^9$ | |
| gold | | | $10^{-1}$ | ±5% |
| silver | | | $10^{-2}$ | ±10% |

To measure a resistor's value you can use a multimeter or the coloured bands. Resistors can have 3,4,5 or 6 bands. Most of the ones we use will have 4 or 5 bands. Usually you will find one group of bands physically separated by another by a small space.

The band on its own is referred to as the tolerance band. In this example its the Gold coloured band. This band determines the range of accuracy of the value measured by the other bands.

The last band(yellow) in the 3 band group is the multiplier and the first 2 bands(red and violet)  form the digits of the resistance value.

# Resistors

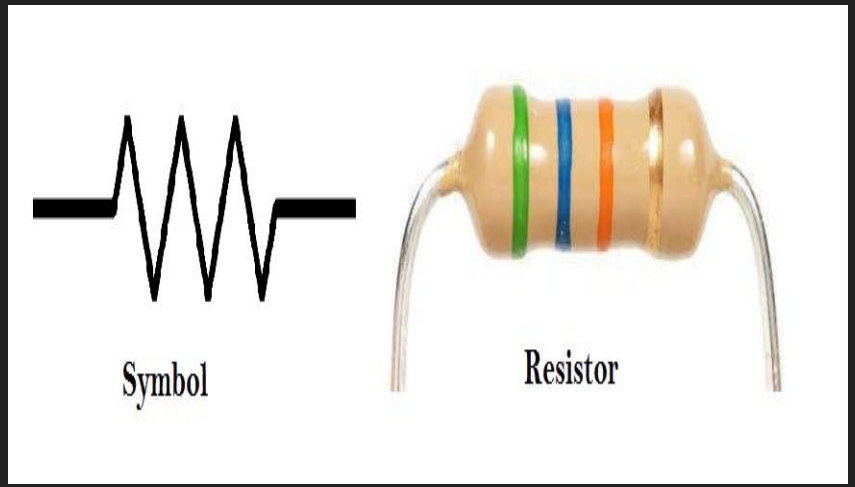


| Color | Digits | | Multiplier | Tolerance |
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| black | 0 | 0 | $10^0$ | |
| brown | 1 | 1 | $10^1$ | ±1% |
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| orange | 3 | 3 | $10^3$ | |
| yellow | 4 | 4 | $10^4$ | |
| green | 5 | 5 | $10^5$ | |
| blue | 6 | 6 | $10^6$ | |
| violet | 7 | 7 | $10^7$ | |
| grey | 8 | 8 | $10^8$ | |
| white | 9 | 9 | $10^9$ | |
| gold | | | $10^{-1}$ | ±5% |
| silver | | | $10^{-2}$ | ±10% |

Here's how we calculate this resistors value:
Red has digit 2
Violet has digit 7
Yellow has the multiplier 4
So the resistor's approximate value is
$27x10^4=27x10000=270,000$ Ohms
270,000 Ohms =270 KOhms
Note that this is it's approximate value. The value of any resistor is not precise but varies by the tolerance value.

The tolerance value is +-5% (Gold band).
So the possible range of values is
270 x 5% = 13.5
270-13.5=256.5
270+13.5=283.5
So the range of possible values is between 256.5 to 283.5 KOhms

# What’s the resistance in this resistor?




Green 5 Blue 6 Orange $10^3$

$56x10^3=56x1000=56,000$ Ohms

=56 KOhms

Tolerance(Gold) 5%

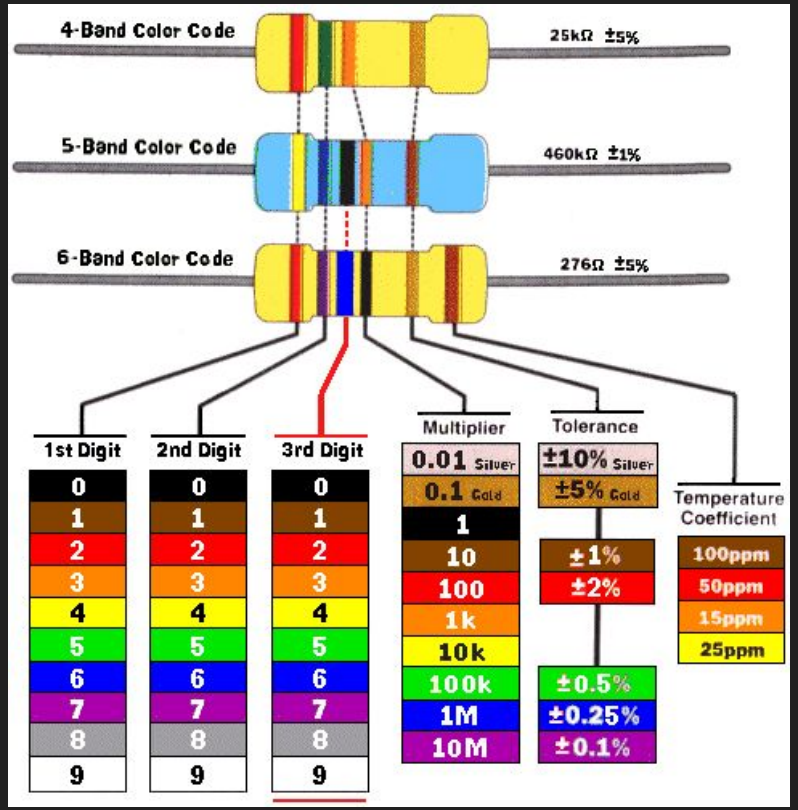
56x5%=2.8

56-2.8=53.2

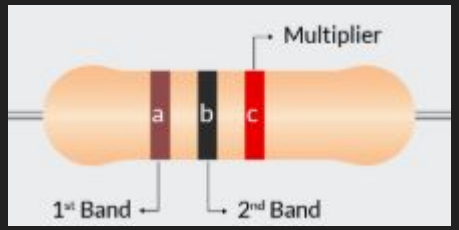
56+2.8=58.8

Range of resistance is 53.2 to 58.8 KOhms

# Resistor Colour Bands




A 3 band resistor has a 20% tolerance rating




In this example the resistors value would be:
$10 \times 10^2 = 1000$ Ohms
Tolerance = 1000 x 20%=200 Ohms
Range is 800 to 1200 Ohms

If you have a 6 band resistor, the last band is a Temperature Coefficient which we will not deal with in this class. The second last band is the tolerance band.

The last band in the large group of bands is always the multiplier band.

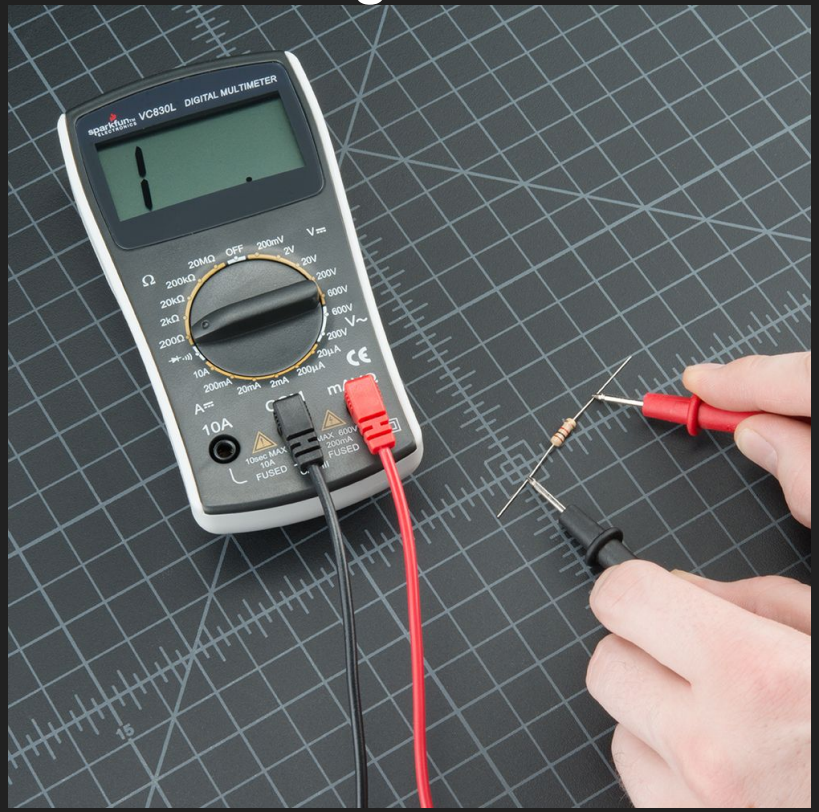
# Multimeter

When using a multimeter make sure to always leave the black probe connected to the COM port and the red one connected to the V/Ω port.

The A or 20A port is only used when measuring current (and we won't be doing that)



# Measuring Resistance Using a Multimeter(MM)




Set MM to Ohms

Move dial to lowest setting

If you see 1 then the setting is too low, so continue to the next level up

Stop when you get a reading

**Multiply by 1000 for K and 1000000 for M**




Watch this video:
https://youtube.com/shorts/NaK43KJVeCs?si=EwCefHatHX0M8J6C

# Continuity Testing

To measure continuity means we are testing if there is a path for electrons to flow between two points on a circuit. This often helps us determine where there are problems. For example if an LED doesn't turn on and everything looks fine it might be because there is a break in the circuit preventing electrons from flowing through it.

To measure continuity on a multimeter find the diode symbol: 

Place your dial to this setting. Touch your two probes together and you should hear a sound. This means there is continuity between the probes. Find a piece of wire and place each probe at the each end of the wire. You should again hear a sound because there is always a path for electrons to flow from one end of a complete wire to the other. Now place the probes on two parts of a table. There should be no sound because electrons cannot flow from the end of one probe through to the other across an insulator like the wood in your table.

Watch this video: https://youtube.com/shorts/jgCzImQaMkA?si=Ba-plinFigsyf9JO



# Questions and Exercises

1. Complete the this worksheet:
   https://docs.google.com/document/d/1w4aZKafPhIge_G0N6oQjzyFxOlS1vlXRcKqER8rk_XA/edit?usp=sharing

   https://docs.google.com/document/d/1XpfLY7Gc3SW3pMYt1qvhU8tatlG13Bnfg7UPLcOxdbM/edit?usp=sharing

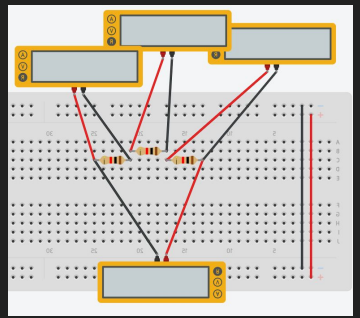
2. Find 3 different physical resistors in class. Take a close up picture of each with your cell phone. Embed the pictures in the next few slides. Next to each indicate their nominal resistance and tolerance. Calculate and display the range of possible values and then finally find the actual resistance using a multimeter.

# Resistor 1

# Resistor 2

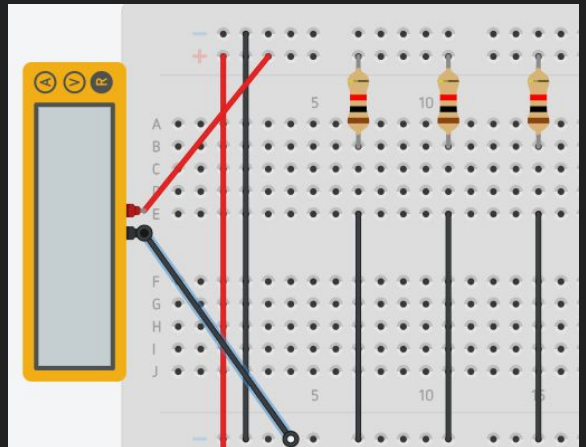
# Resistor 3

#3 Using a breadboard, embed three resistors in series (the tinkercad screenshot is meant to help you). Embed a picture of the circuit below. Using the multimeter measure the resistance across each resistor. Next measure the resistance across all three resistors. What relationship do you notice between these three values?The total resistance of a series resistor is equal to the sum of the individual resistances.



#4 Using a breadboard, embed three resistors in parallel (the tinkercad screenshot is meant to help you). Embed a picture of the circuit below. Using the multimeter measure the resistance across each resistor. Next measure the resistance across all three resistors. What relationship do you notice between these three values?

The total resistance of the parallel resistors is smaller than that of any single resistor.



#5 Measuring continuity means you are measuring the connection between two points in a circuit that have no resistance between them i.e. there is a free flow of electrons between those two points. The multimeter has a continuity setting normally associated with a diode and sound symbol.



Place the dial here and then touch the two probes

together. What happens?The multimeter beeps.

Test for continuity between two ends of a wire. What happens? The multimeter beeps.

Test for continuity between two ends of a resistor. What happens?The multimeter won't make a sound.

Note: power should be off when testing for continuity (you are measuring resistance).

#6 Place a push button into your breadboard as shown in the diagram.

Test for continuity between all four pins (A,B,C and D) on the push button.

Test again when the button is pushed.

Explain your results.

**Measurement Results:**

When the push button is **not pressed**, the multimeter shows **no continuity (infinite resistanc**

When the push button is **pressed**, the multimeter shows **continuity (close to 0 Ω or beeps)**.

**Explanation:**

When the button is not pressed, the internal contacts are **open**, so no current can flow.

When the button is pressed, the internal contacts **close**, creating a path for current to flow.

